Coupled sign and magnitude memory amplifies market-order-flow noise

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Persistent market-order flow is usually diagnosed from order signs, although finite-window fluctuations weight sign pairs by their flow magnitudes. We derive an exact finite-sample decomposition of weekly signed-flow variance into an analytic marginal reference, sign ordering at stratum-homogenised magnitudes, and a sign–magnitude remainder. In four primary cryptocurrency assets, quarterly $\times$ hour-of-week conditioning gives raw-flow variance amplifications of 2.18–3.32. Sign ordering accounts for only 21–35% of this excess; the coupled remainder accounts for 65–79%, with eight-week moving-block 95% lower bounds of 0.52–0.63. Volume normalisation reduces the separation, and monthly conditioning lowers the raw coupling share to 45–49%; with two-point biweekly strata the attribution is structurally unidentifiable. Thus the measured weekly noise is not an additive sum of independent sign and magnitude memories. At the declared conditioning scale, persistent sign runs are predominantly amplified by their joint organisation with flow magnitude.

Introduction. The direction of market orders has long memory: buy and sell signs remain correlated over thousands of events [1, 2]. Order splitting provides a microscopic explanation because a large parent order generates a sequence of child orders with the same sign [3–5]. More recent theories treat persistent signed flow together with fluctuating volume, volatility, and impact [6, 7]. These developments make a simple attribution question unavoidable. When accumulated flow is super-Poissonian, how much of the excess is carried by sign ordering at a typical magnitude, and how much requires signs and magnitudes to remain jointly organised?

This is not answered by comparing their marginal autocorrelations. For a coarse-grained signed flow $x_t = s_t m_t$, every off-diagonal contribution to the variance contains the product $s_t s_{t'} m_t m_{t'}$. Destroying sign order and destroying magnitude order are also asymmetric operations: a sign-randomised series cannot reveal magnitude clustering when the random signs have zero mean. We therefore replace a comparison of surrogate ratios by an additive identity whose reference and remainder are explicit.

The construction borrows the finite-window viewpoint of counting statistics [8–10], but requires no transfer-size quantum and assigns no thermodynamic meaning to market flow. Its output is a conditional variance attribution, not a temperature, entropy production, or causal structural model.

Data and conditioning. Public Binance candles provide total base volume V_t and taker-buy base volume V_t^+. We analyse hourly raw signed volume $\epsilon_t^{(r)} = 2V_t^+ - V_t$ and normalised imbalance $\epsilon_t^{(n)} = \epsilon_t^{(r)}/V_t$ over four years. BTC, ETH, BNB, and SOL form the primary panel; XRP, ADA, DOGE, and AVAX are a post-hoc extension and are not substituted into the original decision rule. Complete Monday–Sunday UTC weeks give $W = 206$ weekly observations.

For a declared calendar partition $g(t)$, we subtract the mean within each period $\times$ UTC hour-of-week stratum and write the residual as $x_t = s_t m_t$. The primary partition is calendar quarter $\times$ hour of week. Monthly and consecutive 14-day partitions test which temporal scales the conditioning absorbs. Let

$$Q_w = \sum_{t\in w} x_t, \qquad V_{\rm obs} = \mathrm{Var}_w(Q_w). \tag{1}$$

Because the residual sums to zero within every stratum, it also sums to zero over the retained sample.

Exact attribution. Every complete week contains at most one observation from any (period, hour-of-week) stratum. Independent permutation of the complete x_t values within strata therefore has zero expected off-diagonal weekly product. Its variance expectation is not a Monte Carlo parameter but the exact diagonal reference

$$V_0 = \frac{1}{W-1}\sum_t x_t^2. \tag{2}$$

This reference preserves the empirical one-point distribution within every stratum.

To define the sign-order contribution on the same calendar footing, let $\widetilde{s}_t = s_t - \langle s\rangle_{g(t)}$ and $\bar{m}_{g(t)} = \langle m\rangle_{g(t)}$. We homogenise only the magnitudes in the off-diagonal term:

$$\begin{aligned} C_s &= \sum_w \sum_{\substack{t,t'\in w\\ t\neq t'}} \widetilde{s}_t \widetilde{s}_{t'} \bar{m}_{g(t)} \bar{m}_{g(t')}, \\ V_s &= \frac{C_s}{W-1}, \qquad V_{sm} = V_{\rm obs} - V_0 - V_s. \end{aligned} \tag{3}$$

Thus $V_{\rm obs} = V_0 + V_s + V_{sm}$ exactly. The measured excess fractions are $f_s = V_s/(V_{\rm obs} - V_0)$ and $f_{sm} = 1 - f_s$; they are not clipped to $[0, 1]$. The observed diagonal remains in V_0, so $V_0 + V_s$ is not the ordinary variance of the homogenised proxy $\widetilde{s}_t \bar{m}_{g(t)}$. Equation (3) declares

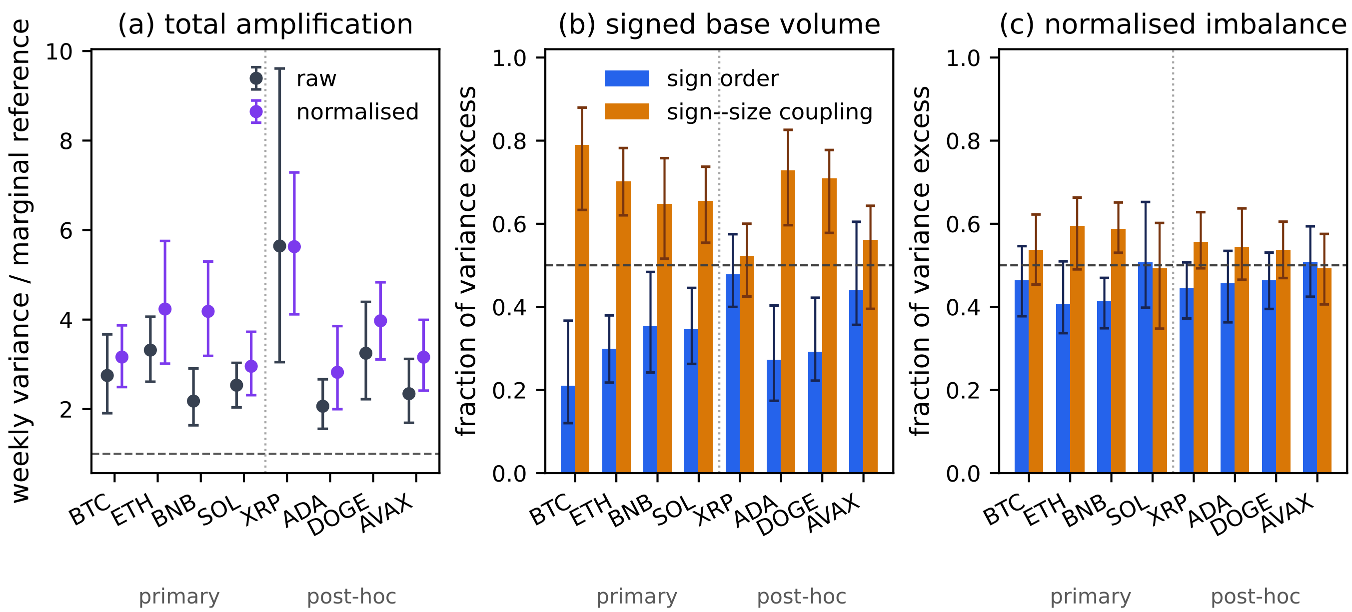

FIG. 1. Weekly amplification and exact attribution under quarterly × hour-of-week conditioning. (a) Total variance relative to the analytic marginal reference. (b),(c) Fractions of the variance excess due to sign order at homogenised magnitudes and the sign–magnitude remainder. Error bars are paired eight-week moving-block 95% intervals. The first four assets are primary; the extension is post hoc.

a fixed-diagonal allocation of the interaction. It is auditable and finite-sample exact, but neither orthogonal nor a unique causal decomposition.

Uncertainty is estimated by paired circular moving-block resampling of the weekly vectors entering all three terms. We use 3999 replicates and block lengths of 4, 8, 13, and 26 weeks. Stratum means are held fixed, so the intervals describe temporal sampling uncertainty conditional on the declared calendar partition. The code reconstructs every Q_w^2 from its three pieces to machine precision [11, 12].

Results. For raw signed volume, $V_{\rm obs}/V_0$ is 2.75, 3.32, 2.18, and 2.53 for BTC, ETH, BNB, and SOL [Fig. 1(a)]. The corresponding sign-order shares are 0.21, 0.30, 0.35, and 0.35. The coupled shares are therefore 0.79, 0.70, 0.65, and 0.66; their eight-week bootstrap intervals are $[0.63, 0.88]$, $[0.62, 0.78]$, $[0.52, 0.76]$, and $[0.55, 0.74]$. The coupled contribution is larger than one half in every primary point estimate and throughout all four intervals.

Normalised imbalance has a larger total amplification, 2.96–4.24, but a less separated attribution: $f_s = 0.41$–0.51 and $f_{sm} = 0.49$–0.59. Normalisation raises the point estimate of f_s in all four primary assets, as expected when broad magnitude variation is removed, but paired intervals resolve that increase clearly only for BTC. We therefore use this direction as a consistency observation, not an independent universal test.

The post-hoc raw-flow coupling shares span 0.52–0.73. Two of their four eight-week intervals cross one half, so the extension supports a non-negligible interaction but not an all-market majority claim. The original predeclared screen was stricter and different: raw observed variance had to exceed twice the 98.75th joint-null percentile in all four primary assets. BNB failed that screen. Neither the normalised analysis nor the exact decomposition rewrites that outcome.

The attribution is scale-conditioned. With monthly × hour-of-week strata the primary raw coupling shares fall to 0.45–0.49, with intervals straddling one half. A 14-day partition leaves at most two values in each stratum; demeaning two values makes their residual magnitudes equal, so magnitude coupling is unidentifiable by construction. We report this degeneracy rather than interpreting a zero remainder as absence of coupling. The contrast between quarterly and monthly conditioning localises much of the measured interaction to multiweek organisation that finer conditioning absorbs.

Discussion. The central result is narrower than "sign memory dominates." Sign persistence is necessary for a positive off-diagonal contribution, but at the primary quarterly conditioning scale it carries only a minority of the raw weekly excess when magnitudes are homogenised. Most of that excess requires the actual temporal pairing of sign runs with flow magnitude. This is consistent with order splitting, where one parent order jointly produces a same-sign sequence and elevated executed flow, but candle data do not identify individual metaorders; the mechanism remains an interpretation rather than a direct account-level measurement.

Two common supporting plots would not add independent evidence. Accumulated variance is fixed by the measured autocorrelation through $\mathrm{Var}(Q_T) = \sum_{t,t'} C_x(t - t')$. Likewise, for a Gaussian Q_T, $\log[P(Q)/P(-Q)] = 2\langle Q_T\rangle Q/\mathrm{Var}(Q_T)$ exactly, so a linear fluctuation symmetry with that slope is an algebraic null, not a separate fluctuation theorem. We keep those checks in the companion analysis and base the present claim only on the finite-sample attribution and its uncertainty [13].

The result therefore concerns a specified observable, horizon, venue, and conditioning scale. Within that boundary, it shows why sign autocorrelation alone is insufficient: finite-window counting noise is controlled by a multiplicative sign–magnitude organisation that an additive reading misses.

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